Mitochondria

Mitochondria are the powerhouses of the cell. They produce ATP through cellular respiration, breaking down glucose in the presence of oxygen. Cells that need a lot of energy, such as muscle cells, contain many mitochondria. Mitochondria have their own small circular DNA, which is inherited from the mother.

Osmosis

Osmosis is the movement of water molecules across a partially permeable membrane, from a region of lower solute concentration to a region of higher solute concentration. A raisin placed in water swells because water enters it by osmosis. Plant cells become turgid when they take in water.

Enzymes

Enzymes are biological catalysts made of protein. Each enzyme has an active site that fits a specific substrate, like a lock and key. Enzymes work best at an optimum temperature; human enzymes work best near 37 °C. High temperatures denature enzymes by changing the shape of the active site.

DNA replication

DNA replication happens before a cell divides. The double helix unzips as the hydrogen bonds between base pairs break. Each strand acts as a template, and free nucleotides pair with it: adenine with thymine and cytosine with guanine. The result is two identical DNA molecules, each with one old and one new strand.

Reflexes

The nervous system carries messages as electrical impulses along neurons. A reflex arc is a fast, automatic response: a receptor detects a stimulus, a sensory neuron carries the impulse to the spinal cord, a relay neuron passes it on, and a motor neuron carries it to an effector such as a muscle.

Photosynthesis

Photosynthesis takes place in chloroplasts. Plants use light energy absorbed by chlorophyll to convert carbon dioxide and water into glucose and oxygen. The rate of photosynthesis is limited by light intensity, carbon dioxide concentration and temperature.